TC-001 - Safety fallback overrides expensive electricity and capacity limits.

Field	Notes
ID	TC-001
Description	Safety fallback overrides expensive electricity and capacity limits.
Traceability	User stories: F3, F5 Requirements: FR3, FR5, NFR9 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated. Safety margin is 2.0 C.
Test Data	HeatPumpData: HP1, current 17.0, target 21.0, buffer 40.0, running false. Price: expensive. currentlyRunning: 99.
Steps	1. Create HeatPumpData. 2. Call decide(data, expensive, 99).
For each step	
Expected Result	START_HEATING is returned because the heat deficit is above the safety fallback margin.
Actual Result	START_HEATING was returned.
Pass/Fail	**PASS**

TC-002 - Safety fallback activates exactly at the defined margin.

Field	Notes
ID	TC-002
Description	Safety fallback activates exactly at the defined margin.
Traceability	User stories: F5 Requirements: FR5, NFR9 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 19.0, target 21.0. Price: expensive. currentlyRunning: 99.
Steps	1. Create HeatPumpData. 2. Call decide().
For each step	
Expected Result	START_HEATING is returned when deficit equals 2.0 C.
Actual Result	START_HEATING was returned.
Pass/Fail	**PASS**

TC-003 - Cheap electricity and heat demand starts heating when capacity is available.

Field	Notes
ID	TC-003
Description	Cheap electricity and heat demand starts heating when capacity is available.
Traceability	User stories: F3, F4 Requirements: FR3, FR4 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 20.0, target 21.0. Price: cheap. currentlyRunning: 0.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	START_HEATING is returned.
Actual Result	START_HEATING was returned.
Pass/Fail	**PASS**

TC-004 - Cheap electricity stores heat in buffer when house is already warm.

Field	Notes
ID	TC-004
Description	Cheap electricity stores heat in buffer when house is already warm.
Traceability	User stories: F8 Requirements: FR8 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 21.5, target 21.0, buffer 40.0. Price: cheap. currentlyRunning: 0.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	STORE_HEAT_IN_BUFFER is returned.
Actual Result	STORE_HEAT_IN_BUFFER was returned.
Pass/Fail	**PASS**

TC-005 - Full buffer prevents additional buffer storage.

Field	Notes
ID	TC-005
Description	Full buffer prevents additional buffer storage.
Traceability	User stories: F8 Requirements: FR8 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 21.5, target 21.0, buffer 55.0. Price: cheap.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	Command is not STORE_HEAT_IN_BUFFER because the buffer is full.
Actual Result	STORE_HEAT_IN_BUFFER was not returned.
Pass/Fail	**PASS**

TC-006 - Cheap electricity with heat demand waits when capacity is full.

Field	Notes
ID	TC-006
Description	Cheap electricity with heat demand waits when capacity is full.
Traceability	User stories: F6 Requirements: FR6 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 20.0, target 21.0. currentlyRunning equals MAX_RUNNING_HEAT_PUMPS.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	WAIT is returned because capacity is unavailable.
Actual Result	WAIT was returned.
Pass/Fail	**PASS**

TC-007 - Cheap electricity and buffer opportunity waits when capacity is full.

Field	Notes
ID	TC-007
Description	Cheap electricity and buffer opportunity waits when capacity is full.
Traceability	User stories: F6, F8 Requirements: FR6, FR8 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 21.5, target 21.0, buffer 40.0. currentlyRunning equals MAX_RUNNING_HEAT_PUMPS.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	WAIT is returned because storing heat also requires capacity.
Actual Result	WAIT was returned.
Pass/Fail	**PASS**

TC-008 - Medium price with small heat demand waits.

Field	Notes
ID	TC-008
Description	Medium price with small heat demand waits.
Traceability	User stories: F3, F4 Requirements: FR3, FR4 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 20.0, target 21.0. Price: medium.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	WAIT is returned because electricity is not cheap and safety fallback is not needed.
Actual Result	WAIT was returned.
Pass/Fail	**PASS**

TC-009 - Medium price and warm house waits.

Field	Notes
ID	TC-009
Description	Medium price and warm house waits.
Traceability	User stories: F3, F4 Requirements: FR3, FR4 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 21.5, target 21.0. Price: medium.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	WAIT is returned.
Actual Result	WAIT was returned.
Pass/Fail	**PASS**

TC-010 - Expensive electricity stops a running pump when the house is warm.

Field	Notes
ID	TC-010
Description	Expensive electricity stops a running pump when the house is warm.
Traceability	User stories: F3, F4 Requirements: FR3, FR4 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 21.5, target 21.0, running true. Price: expensive.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	STOP_HEATING is returned.
Actual Result	STOP_HEATING was returned.
Pass/Fail	**PASS**

TC-011 - Expensive electricity makes an idle warm house wait.

Field	Notes
ID	TC-011
Description	Expensive electricity makes an idle warm house wait.
Traceability	User stories: F3, F4 Requirements: FR3, FR4 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 21.5, target 21.0, running false. Price: expensive.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	WAIT is returned.
Actual Result	WAIT was returned.
Pass/Fail	**PASS**

TC-012 - Expensive electricity and small deficit waits when safety fallback is not needed.

Field	Notes
ID	TC-012
Description	Expensive electricity and small deficit waits when safety fallback is not needed.
Traceability	User stories: F3, F4, F5 Requirements: FR3, FR4, FR5 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HeatPumpData: current 20.0, target 21.0. Price: expensive.
Steps	1. Create data. 2. Call decide().
For each step	
Expected Result	WAIT is returned.
Actual Result	WAIT was returned.
Pass/Fail	**PASS**

TC-013 - Capacity prioritisation starts pumps with largest heat deficit first.

Field	Notes
ID	TC-013
Description	Capacity prioritisation starts pumps with largest heat deficit first.
Traceability	User stories: F6, F7 Requirements: FR6, FR7 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	HP-A deficit 1.0, HP-B deficit 3.5, HP-C deficit 1.5. Price: cheap.
Steps	1. Create list of three clients. 2. Call decideAll().
For each step	
Expected Result	HP-B and HP-C receive START_HEATING. HP-A receives WAIT.
Actual Result	Expected commands were returned.
Pass/Fail	**PASS**

TC-014 - No connected clients returns an empty decision map.

Field	Notes
ID	TC-014
Description	No connected clients returns an empty decision map.
Traceability	User stories: F3 Requirements: FR3 Use cases: UC3
Assumptions and Pre-Conditions	ControlDecisionService is instantiated.
Test Data	Empty client list. Price: cheap.
Steps	1. Call decideAll(empty list, cheap).
For each step	
Expected Result	Returned map is empty.
Actual Result	Returned map was empty.
Pass/Fail	**PASS**

TC-015 - ElectricityPriceReader loads multiple monthly prices correctly.

Field	Notes
ID	TC-015
Description	ElectricityPriceReader loads multiple monthly prices correctly.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC1, UC3
Assumptions and Pre-Conditions	Temporary CSV file is available.
Test Data	Rows: Januar 73.09, Februar 84.18, Maj 47.92 øre/kWh.
Steps	1. Write CSV. 2. Call readPrices().
For each step	
Expected Result	Three entries are loaded and converted from øre/kWh to DKK/kWh.
Actual Result	Three entries were loaded with expected values.
Pass/Fail	**PASS**

TC-016 - ElectricityPriceReader converts øre/kWh to DKK/kWh.

Field	Notes
ID	TC-016
Description	ElectricityPriceReader converts øre/kWh to DKK/kWh.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC1, UC3
Assumptions and Pre-Conditions	Temporary CSV file is available.
Test Data	Juni price: 43.85 øre/kWh.
Steps	1. Write CSV. 2. Call readPrices().
For each step	
Expected Result	Juni is returned as 0.4385 DKK/kWh.
Actual Result	0.4385 was returned.
Pass/Fail	**PASS**

TC-017 - ElectricityPriceReader skips annual average rows.

Field	Notes
ID	TC-017
Description	ElectricityPriceReader skips annual average rows.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC1
Assumptions and Pre-Conditions	Temporary CSV file is available.
Test Data	Rows: Januar 73.09 and Aarsgennemsnit 60.67.
Steps	1. Write CSV. 2. Call readPrices().
For each step	
Expected Result	Only Januar is included; Aarsgennemsnit is skipped.
Actual Result	Only Januar was included.
Pass/Fail	**PASS**

TC-018 - ElectricityPriceReader does not parse the header row as data.

Field	Notes
ID	TC-018
Description	ElectricityPriceReader does not parse the header row as data.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC1
Assumptions and Pre-Conditions	Temporary CSV file is available.
Test Data	Header row and one Januar row.
Steps	1. Write CSV. 2. Call readPrices().
For each step	
Expected Result	Header key is not included in the result map.
Actual Result	Header key was not included.
Pass/Fail	**PASS**

TC-019 - ElectricityPriceReader returns an empty map when the CSV file is missing.

Field	Notes
ID	TC-019
Description	ElectricityPriceReader returns an empty map when the CSV file is missing.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC1
Assumptions and Pre-Conditions	The file path does not exist.
Test Data	Path: does_not_exist.csv.
Steps	1. Call readPrices() with missing path.
For each step	
Expected Result	An empty map is returned.
Actual Result	Empty map was returned.
Pass/Fail	**PASS**

TC-020 - Price service uses January at startup.

Field	Notes
ID	TC-020
Description	Price service uses January at startup.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC1, UC5
Assumptions and Pre-Conditions	ElectricityPriceService is instantiated.
Test Data	Price map: Januar = 0.7309.
Steps	1. Create service. 2. Call getCurrentMonthName() and getCurrentPrice().
For each step	
Expected Result	Current month is Januar and current price is 0.7309.
Actual Result	Expected month and price were returned.
Pass/Fail	**PASS**

TC-021 - Price service falls back to default when the current month is missing.

Field	Notes
ID	TC-021
Description	Price service falls back to default when the current month is missing.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC1, UC5
Assumptions and Pre-Conditions	ElectricityPriceService is instantiated.
Test Data	Price map contains Maj only.
Steps	1. Create service. 2. Call getCurrentPrice().
For each step	
Expected Result	Default price 0.61 is returned for missing Januar.
Actual Result	0.61 was returned.
Pass/Fail	**PASS**

TC-022 - Price service returns a configured price by key.

Field	Notes
ID	TC-022
Description	Price service returns a configured price by key.
Traceability	User stories: F4 Requirements: FR4, NFR8 Use cases: UC3
Assumptions and Pre-Conditions	ElectricityPriceService is instantiated.
Test Data	Price map: Maj = 0.4792.
Steps	1. Create service. 2. Call getPriceForKey('Maj').
For each step	
Expected Result	0.4792 is returned.
Actual Result	0.4792 was returned.
Pass/Fail	**PASS**

TC-023 - Price service identifies cheap electricity at threshold.

Field	Notes
ID	TC-023
Description	Price service identifies cheap electricity at threshold.
Traceability	User stories: F3, F4 Requirements: FR3, FR4 Use cases: UC3
Assumptions and Pre-Conditions	ElectricityPriceService is instantiated.
Test Data	Price map: Januar = 0.55.
Steps	1. Create service. 2. Call isCheap() and isExpensive().
For each step	
Expected Result	isCheap is true and isExpensive is false.
Actual Result	Expected boolean values were returned.
Pass/Fail	**PASS**

TC-024 - Price service identifies expensive electricity at threshold.

Field	Notes
ID	TC-024
Description	Price service identifies expensive electricity at threshold.
Traceability	User stories: F3, F4 Requirements: FR3, FR4 Use cases: UC3
Assumptions and Pre-Conditions	ElectricityPriceService is instantiated.
Test Data	Price map: Januar = 0.75.
Steps	1. Create service. 2. Call isExpensive() and isCheap().
For each step	
Expected Result	isExpensive is true and isCheap is false.
Actual Result	Expected boolean values were returned.
Pass/Fail	**PASS**

TC-025 - HeatPumpData parses a valid client status message.

Field	Notes
ID	TC-025
Description	HeatPumpData parses a valid client status message.
Traceability	User stories: F2, F9 Requirements: FR2, FR9 Use cases: UC2, UC3, UC5
Assumptions and Pre-Conditions	Raw message follows the defined protocol.
Test Data	CLIENT_ID=HP1;CURRENT_TEMP=19.2;TARGET_TEMP=21.0;BUFFER_TEMP=42.0;RUNNING=false.
Steps	1. Call HeatPumpData.parse(raw). 2. Read all fields.
For each step	
Expected Result	All fields are parsed into the correct values.
Actual Result	All fields matched expected values.
Pass/Fail	**PASS**

TC-026 - HeatPumpData parses running=true correctly.

Field	Notes
ID	TC-026
Description	HeatPumpData parses running=true correctly.
Traceability	User stories: F2, F9 Requirements: FR2, FR9 Use cases: UC3, UC5
Assumptions and Pre-Conditions	Raw message follows the defined protocol.
Test Data	CLIENT_ID=HP2;...;RUNNING=true.
Steps	1. Call parse(). 2. Check isRunning().
For each step	
Expected Result	isRunning returns true.
Actual Result	isRunning returned true.
Pass/Fail	**PASS**

TC-027 - Heat deficit is positive when the house is cold.

Field	Notes
ID	TC-027
Description	Heat deficit is positive when the house is cold.
Traceability	User stories: F2, F3, F5 Requirements: FR2, FR3, FR5 Use cases: UC3, UC5
Assumptions and Pre-Conditions	HeatPumpData object is available.
Test Data	Current 18.0, target 21.0.
Steps	1. Create data. 2. Call getHeatDeficit().
For each step	
Expected Result	Heat deficit is 3.0.
Actual Result	3.0 was returned.
Pass/Fail	**PASS**

TC-028 - Heat deficit is negative when the house is warmer than target.

Field	Notes
ID	TC-028
Description	Heat deficit is negative when the house is warmer than target.
Traceability	User stories: F2, F3 Requirements: FR2, FR3 Use cases: UC3, UC5
Assumptions and Pre-Conditions	HeatPumpData object is available.
Test Data	Current 22.0, target 21.0.
Steps	1. Create data. 2. Call getHeatDeficit().
For each step	
Expected Result	Heat deficit is -1.0.
Actual Result	-1.0 was returned.
Pass/Fail	**PASS**

TC-029 - Null or blank heat pump messages are rejected safely.

Field	Notes
ID	TC-029
Description	Null or blank heat pump messages are rejected safely.
Traceability	User stories: F9 Requirements: FR9, NFR4 Use cases: UC3
Assumptions and Pre-Conditions	No valid message is available.
Test Data	Input: null and blank string.
Steps	1. Call parse(null). 2. Call parse(blank).
For each step	
Expected Result	Both calls return null.
Actual Result	Both calls returned null.
Pass/Fail	**PASS**

TC-030 - HeatPumpData can serialize and parse a round trip.

Field	Notes
ID	TC-030
Description	HeatPumpData can serialize and parse a round trip.
Traceability	User stories: F2, F9 Requirements: FR2, FR9 Use cases: UC2, UC3, UC5
Assumptions and Pre-Conditions	HeatPumpData object is available.
Test Data	HP3, current 20.5, target 21.0, buffer 45.0, running true.
Steps	1. Serialize object. 2. Parse serialized string. 3. Compare fields.
For each step	
Expected Result	Parsed object matches the original object.
Actual Result	All compared fields matched.
Pass/Fail	**PASS**